

Registration Page Wireframe & Layout Specification

50/50 Split-Screen Spatial Trust & Registration Architecture · MacBook Air 16:10 Frame (1440px)

Jira Reference: SCRUM-66

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Design Standards: Balsamiq/Figma Grayscale Blueprint Conventions, Apple HIG Ergonomics, NIST SP 800-63B Authentication, WCAG 2.2 AA Contrast & Touch Rules

Framework: Next.js 16 + React 19 + Supabase Auth

Reviewed By: Hermar Centillas (Lead / Scrum Master)

Fidelity: MID-FIDELITY WIREFRAME

Status: APPROVED / ACTIVE

1. Executive Summary & Map-First Authentication Architecture

AbangCebu AI replaces traditional brochure rental portals with an authentic map-first discovery interface. The landing page IS the map—exploring verified rental rooms across Metro Cebu (CIT-U, USC, IT Park, Urgello, Banilad) begins directly on a vector canvas without forced registration walls. Registration is strictly deferred until high-intent actions (saving favorites, initiating real-time landlord inquiries, listing rooms, or ocular scheduling).

Spatial State Preservation Rule: When an unauthenticated seeker initiates a gated action, current coordinates, zoom level, and selected listing ID are preserved via URL parameter (/register?returnTo=%2Fmap%3Flat%3D10.2942%26lng%3D123.8659%26zoom%3D15%26listing%3Dcit-dorm-04). Upon completing registration and confirmation, the user returns seamlessly to their exact spatial context.

2. Desktop Viewport Specification (1440px × 900px · MacBook Air Chassis)

Encased in an authentic MacBook Air chassis (16:10 ratio), desktop registration implements a 50/50 split-screen architecture. The left pane grounds the seeker in Metro Cebu spatial trust with live listing counts, road network arteries, and dynamic role value propositions (Renter vs Landlord). The right pane presents an elevated 460px maximum-width registration card with NIST SP 800-63B password guidance and Philippine mobile validation.

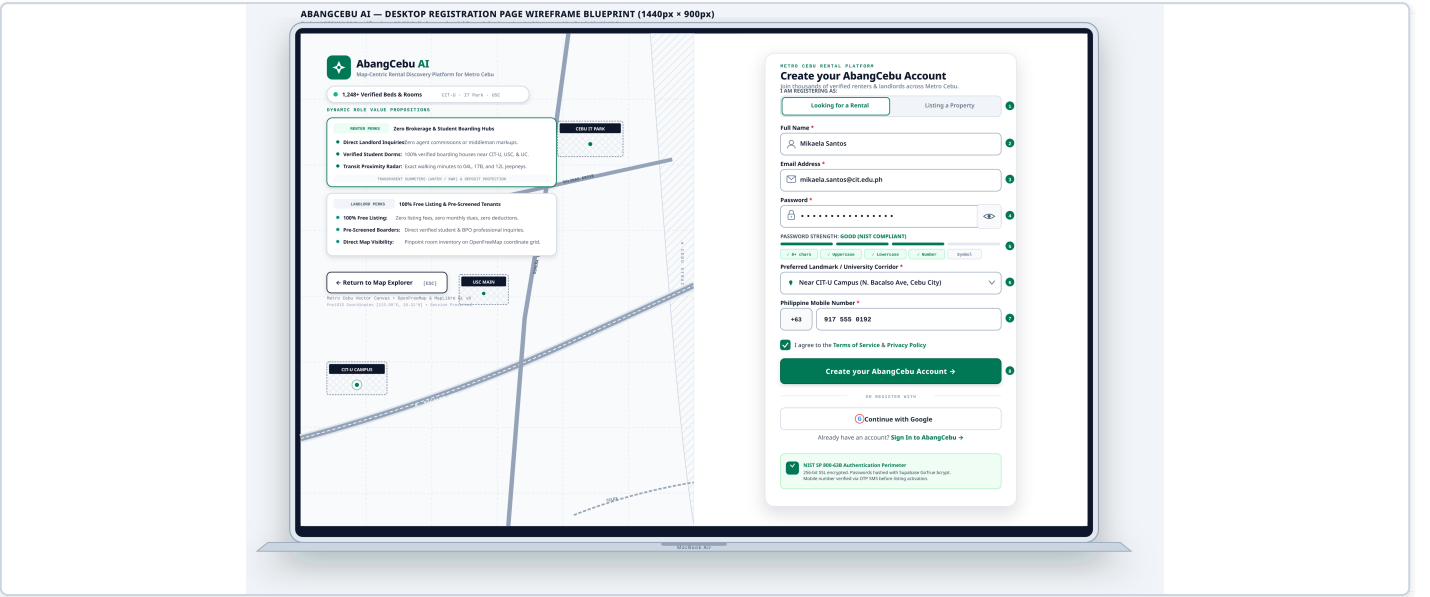


Figure 1: Desktop 1440px Mid-Fidelity Wireframe Blueprint — MacBook Air Aluminum Chassis, 50/50 Split-Screen, Spatial Trust & Metro Cebu Map Grid, 460px Centered Registration Card

Component	Dimensions / Placement	Wireframe Role & Authentic Design Architecture
MacBook Air Chassis	1480 × 950px (Outer)	Aluminum unibody chassis, top FaceTime camera notch, inner matte bezel, and bottom display thumb lip.
Left Spatial Trust Pane	w: 720px, h: 900px (x: 0, y: 0)	Metro Cebu road arteries (N. Bacalso Ave, Osmeña Blvd, Salinas Dr, CCLEX), anchor nodes (CIT-U, IT Park, USC), and Cebu Strait hatch.
Dynamic Role Perks Stack	w: 420px (x: 48, y: 142)	Dynamic presentation: Renter Perks (Zero Brokerage, Verified student boarding near CIT-U/USC, 04L/17B transit) vs Landlord Perks (100% Free Listing, Pre-screened tenants).
Return to Map Trigger	w: 220px, h: 38px (x: 48, y: 444)	Secondary button with ← icon and keyboard shortcut hint [ESC]. Preserves search parameters.
Right Registration Card	w: 460px, h: 828px (x: 850, y: 36)	Elevated 460px card with 28px internal padding, subtle drop shadow, and clean vertical visual rhythm.
Role-Segmented Control ①	w: 404px, h: 38px	Accessible segmented tablist (Looking for Room / Landlord). Active tab features emerald border, bold text, ARIA role="tablist".
Full Name & Email Inputs ②③	w: 404px, h: 40px	Touch-optimized fields with leading vector icons (User, Mail), 16px font rule, and duplicate email detection.
Password & NIST Meter ④⑤	w: 404px, h: 40px + 36px meter	Masked password with 44×44px eye toggle, 4-segment strength gauge, and 5 interactive requirement pills (8+, upper, lower, num, symbol).
Conditional Dropdown & Mobile ⑥⑦	w: 404px, h: 40px each	Role-conditional dropdown (Preferred University Corridor vs Municipality) and Philippine Mobile (+63 prefix, 10-digit number).
Primary CTA & Google OAuth ⑧	w: 404px, h: 46px & 40px	Cebu Emerald (#047857) "Create Account" CTA (5.1:1 contrast) and official Google OAuth PKCE button.

3. Mobile Viewport Geometry & Touch Ergonomics

The mobile viewport standardizes on 100dvh (Dynamic Viewport Height) encased in an authentic iPhone 16 Pro Titanium chassis to prevent iOS Safari address bar clipping. It enforces strict 44×44px touch targets and the 16px input font rule to eliminate destructive auto-zooming. Overlaid below is the side-by-side comparative blueprint across all 4 interactive states:

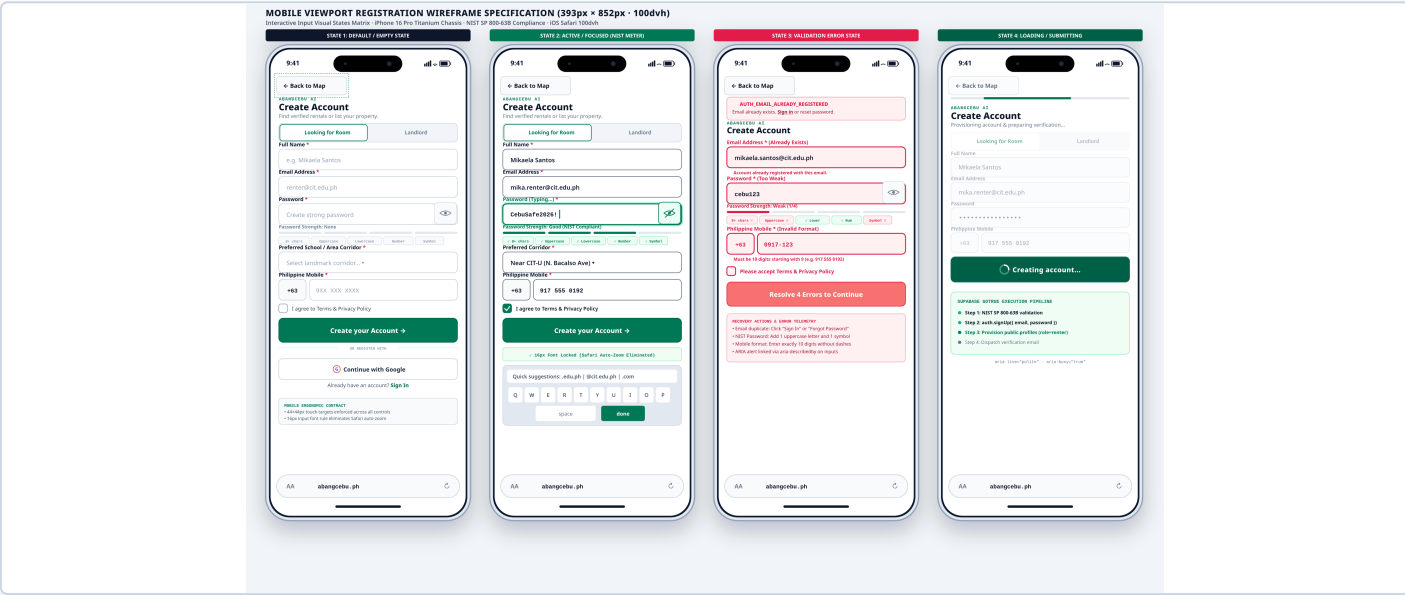


Figure 2: Mobile 100dvh Mid-Fidelity Wireframe Blueprint — iPhone 16 Pro Titanium Chassis, Dynamic Island, State 1 (Default/Empty), State 2 (Active/Focused), State 3 (Validation Error), and State 4 (Loading/Submitting)

Visual State	Visual Attributes	User Experience & Error Contracts	Ergonomic & Technical Governance
State 1: Default / Empty	Neutral borders (#CBD5E1), gray placeholders (#94A3B8), 4 empty gauge segments, 5 gray requirement pills.	Presents clear Renter/Landlord role tabs, empty inputs, Landmark corridor dropdown, and +63 mobile prompt.	Strict 44×44px touch targets; ← Back to Map top safe-area header action button.
State 2: Active / Focused	Emerald focus ring (#047857, 2px), 3/4 filled gauge (Good), 5 active emerald requirement pills, eye-slash pressed.	Plaintext password revealed upon toggle tap (CebuSafe2026!); NIST validation provides immediate real-time feedback.	Input font-size locked to 16px (font-size: 16px !important) to eliminate Safari destructive auto-zoom.
State 3: Validation Error	Crimson border (#E11D48), light red background (#FFF1F2), top duplicate email alert banner.	Displays AUTH_EMAIL_ALREADY_REGISTERED banner + inline errors for weak password, invalid phone, and unchecked terms.	Linked with aria-describedby; screen reader alert announced via role="alert" and aria-live="assertive".
State 4: Loading / Submitting	Disabled form controls (opacity-50), rotating vector spinner inside CTA button, top indeterminate bar.	Button label changes to "Creating account...", click events disabled to prevent double-submission.	Server Action execution with Supabase GoTrue; profile provisioning (role=renter) and verification email dispatch.

MOBILE TOUCH ERGONOMICS & SAFETY GUARDRAILS

- WCAG 2.2 AA 44×44px Touch Targets: Enforced via @utility touch-target on all buttons, segmented tabs, eye toggles, dropdowns, and checkboxes.
- Anti-Zoom 16px Rule: Mobile input font-size strictly locked to 16px (text-base) to eliminate iOS Safari destructive auto-zoom.
- Thumb Zone Optimization: Role tabs, input fields, password gauge, and primary "Create Account" CTA are centered in the lower two-thirds natural reach zone.
- Dynamic Viewport Lock: Root container uses 100dvh overscroll-none preventing rubber-band displacement while interacting.

4. Sequential Keyboard Tab Order & WAI-ARIA Matrix

Tab #	Element	Accessible Name	ARIA Roles & States	Keyboard Action
1	← Back to Map	"Back to Map Explorer"	role="link"	Enter / Space
2	Looking for Room Tab	"Looking for a Rental context"	role="tab", aria-selected="true"	ArrowRight to Landlord
3	Listing Property Tab	"Listing a Property context"	role="tab", aria-selected="false"	ArrowLeft to Renter
4	Full Name Input	"Full Name, required"	type="text", aria-required="true"	Standard typing; Tab
5	Email Address Input	"Email Address, required"	type="email", aria-required="true"	Standard typing; Tab
6	Password Input	"Password, required"	type="password", aria-describedby="pass-meter"	Standard typing
7	Show/Hide Eye Toggle	"Toggle password visibility"	role="button", aria-pressed="false"	Enter / Space toggles
—	NIST Strength Meter	"Password Strength: Good"	role="progressbar", aria-valuenow="75"	Polite live announcement
8	Context Dropdown	"Preferred Landmark / Corridor"	role="combobox", aria-expanded="false"	Alt + Down opens options
9	Country Prefix Pill	"Philippine Country Code +63"	role="button", tabindex="-1"	Locked default
10	Mobile Phone Input	"Mobile number, required"	type="tel", inputmode="numeric"	Numeric typing; Tab
11	Terms Checkbox	"I agree to Terms & Privacy"	role="checkbox", aria-checked="false"	Space toggles check
12	Create Account CTA	"Create your AbangCebu Account"	type="submit", aria-busy="false"	Enter / Space submits
13	Google OAuth Button	"Continue with Google"	role="button"	Enter triggers OAuth
14	Sign In Link	"Sign In to AbangCebu"	role="link"	Enter routes to /login

5. Next.js 16 Component Hierarchy & Boundaries (Sprint 2 Contract)

Component	Runtime	File Location (Sprint 2)	Architectural Role & Contract
RegisterPage	SERVER	src/app/(auth)/register/page.tsx	Validates searchParams.returnTo open-redirect allowlist. Streams metadata & wraps form.
RegisterForm	CLIENT	src/components/auth/register-form.tsx	React 19 useActionState form container managing input states, pending transitions, and focus rings.
RoleSelector	CLIENT	src/components/auth/role-selector.tsx	Accessible WAI-ARIA tablist segmented control switching between Renter and Landlord identity modes.
PasswordStrengthMeter	CLIENT	src/components/auth/password-strength-meter.tsx	Evaluates NIST SP 800-63B criteria in real time. Renders 4-segment gauge and 5 reactive requirement pills.
ConditionalContextDropdown	CLIENT	src/components/auth/conditional-context-dropdown.tsx	Renders university corridor selection for Renters (CIT-U, USC) or municipality for Landlords.
PhilippineMobileInput	CLIENT	src/components/auth/philippine-mobile-input.tsx	Formatted 10-digit mobile field with +63 country code lock and telecom prefix validation.
SocialAuthButtons	CLIENT	src/components/auth/social-auth-buttons.tsx	Executes Google OAuth 2.0 PKCE authorization flow via supabase.auth.signInWithOAuth().
AuthAlertBanner	CLIENT	src/components/auth/auth-alert-banner.tsx	Renders accessible standardized AuthErrorResponse payloads (SCRUM-60) with recovery hints.

6. Definition of Done & Architectural Sign-Off

- Mid-Fidelity Wireframe Standard: Balsamiq/Figma grayscale blueprints, crossed [X] placeholders, and authentic MacBook Air / iPhone 16 Pro chassis.

Desktop Specification: 1440×900px, 50/50 split layout, Metro Cebu spatial map network, dynamic dual perks, and centered 460px registration card.

Mobile 4-State Specification: 100dvh viewport, State 1 (Default), State 2 (Active/ NIST), State 3 (Validation Error), and State 4 (Loading/Submitting).

NIST SP 800-63B Compliance: 4-segment visual gauge bar + 5 interactive requirement pills (8+, uppercase, lowercase, number, symbol).

Accessibility Compliance: WCAG 2.2 AA 44×44px touch targets, 16px mobile input anti-zoom rule, contrast >= 4.5:1, and sequential tab order flow.

Sprint 1 Boundary Maintained: Purely architectural wireframe blueprints and data contracts; ZERO premature React/JSX feature code.

Rendered Engineering Assets: High-res vector SVGs, 300 DPI PNG blueprints, and 3-page WeasyPrint engineering specification PDF.

ARCHITECTURAL SIGN-OFF

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Reviewer: Hermar Centillas (Lead / Scrum Master)

Jira Ticket: SCRUM-66 | Status: Approved & Audited

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